



Department of CSE, PES University

UE24CS341A: SOFTWARE ENGINEERING

5TH SEMESTER, ACADEMIC YEAR 2026-27

LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

DATE: 09-09-2026

NAME: PRATHAM KUMAR

SRN NO.: PES2UG24CS368

PRN NO.: PES2202400086

ASSIGNMENT WORK

Architecture Justification

Podcast Guest Scheduling & Outline Builder

Selected Architecture: Layered Architecture

Architecture Selection

We chose **Layered Architecture** for the **Podcast Guest Scheduling & Outline Builder** because it provides a clear separation between the user interface, application logic, and data management. The system can be organized into a Presentation Layer, Business Layer, and Data Layer, making the different responsibilities easier to manage and maintain.

Reason 1 — Clear Separation of Responsibilities

The podcast system contains different functions such as viewing and booking interview slots, managing host availability, submitting guest information, and generating episode outlines. Layered Architecture separates these responsibilities into appropriate layers. The Presentation Layer handles interaction with the Podcast Guest and Show Host, the Business Layer handles scheduling, guest information, and outline generation, while the Data Layer manages persistent podcast data. This makes the system easier to understand and maintain.

Reason 2 — Suitable for the System Scope

The Podcast Guest Scheduling & Outline Builder has a focused set of related functions and does not require independently deployable services for every function. A layered structure provides the required organization without introducing the additional operational complexity and network communication associated with Microservices Architecture. It also keeps scheduling and booking information within a consistent application structure.

Security Advantage

Layered Architecture allows authorization checks to be handled before host-specific operations reach the data layer. This helps restrict functions such as creating, updating, or blocking interview slots and managing episode information to authorized users. This supports **NFR-002**, which requires host-specific scheduling and episode-management functions to be accessible only to authorized users.

Performance Benefit

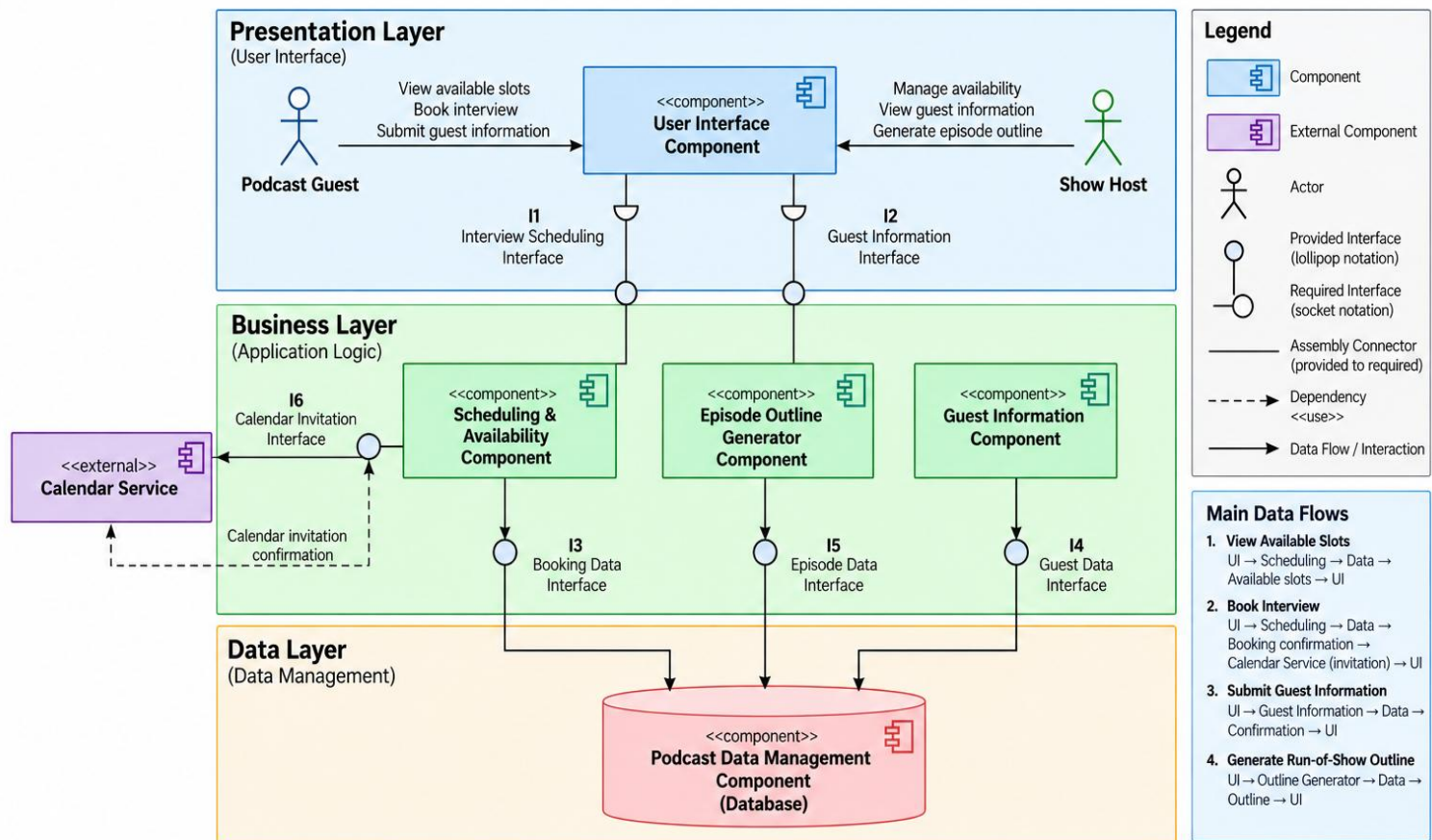
The layered structure keeps the main application processing within the system and avoids the additional network communication that can occur between independently deployed microservices. This can help reduce unnecessary communication overhead for operations such as generating the production sheet. This supports **NFR-001**, which requires the formatted PDF production sheet to be generated in under one second.

Conclusion

Therefore, **Layered Architecture** is the most appropriate architecture for the **Podcast Guest Scheduling & Outline Builder** because it provides clear separation of responsibilities, is suitable for the current system scope, supports controlled access to host functions, and helps maintain efficient processing for production-sheet generation.

Podcast Guest Scheduling & Outline Builder

UML Component Diagram (Layered Architecture)





Department of CSE, PES University

UE24CS341A: SOFTWARE ENGINEERING

5TH SEMESTER, ACADEMIC YEAR 2026-27

LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

DATE: 09-09-2026

NAME: PRATHAM KUMAR

SRN NO.: PES2UG24CS368

PRN NO.: PES2202400086

SECTION: F

THANK YOU